

## Chapter 14 Chemical Kinetics

### Review

- **Chemical kinetics:** the area of chemistry concerned with the speeds or rates of reactions.
- **Reaction rate:** the speed at which a chemical reaction occurs / the change in the concentration of reactants or products per unit of time.
- **The collision model:** molecules must collide to react, the greater the number of collisions per second, the greater the reaction rate.
- **Factors that affect reaction rate:** the greater the frequency of collisions, the higher the reaction rate.
  - 1) Physical state of the reactant; *heterogeneous; rate  $\propto$  area of contact*
  - 2) Reactant concentration;
  - 3) Reaction temperature;  *$T \uparrow \rightarrow$  collision  $\uparrow$ , rate  $\uparrow$*
  - 4) The presence of a catalyst;
- **Instantaneous rate:** the rate at a particular instant during the reaction, it is determined from the slope of the curve at a particular point in time.
- **Reaction rates and stoichiometry:**

$$aA + bB \rightarrow cC + dD$$

$$\text{rate} = -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}$$

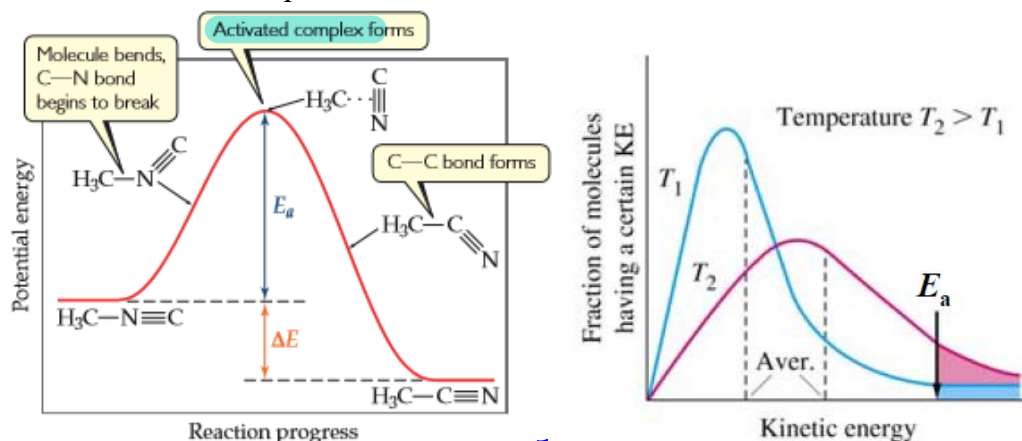
- **Rate law:** the equation shows how the rate depends on reactant concentrations. *measured experimentally*

$$aA + bB \rightarrow cC + dD$$

$$\text{rate} = k[A]^m[B]^n$$
  - 1)  $k$  is the rate constant, the magnitude of  $k$  is affected by temperature and by the presence of a catalyst, the unit of rate constant depend on the overall reaction order of the rate law.
  - 2)  $m$  and  $n$  is the reaction order.
  - 3) For any reaction, the rate law must be determined experimentally.
  - 4) A large value of  $k$  ( $\sim 10^9$  or higher) means a fast reaction and a small value of  $k$  (10 or lower) means a slow reaction.
- **Overall reaction order:** the sum of the orders with respect to each reactant represented in the rate law.
- **First-order reaction:** rate depends on the concentration of a single reactant raised to the first power.
- **Second-order reaction:** rate depends either on a reactant concentration raised to the second power or on the concentrations of two reactants each raised to the first power.
- **Zero-order reaction:** rate of disappearance of A is independent of [A].
- **Half-life ( $t_{1/2}$ ):** the time required for the concentration of a reactant to reach half its initial value.
- **Summary:**

| Reaction order | Differential rate law       | Integrated rate law                         | Half-life ( $t_{1/2}$ ) |
|----------------|-----------------------------|---------------------------------------------|-------------------------|
| <b>0</b>       | $-\frac{d[A]}{dt} = k$      | $[A] = [A]_0 - kt$                          | $\frac{[A]_0}{2k}$      |
| <b>1</b>       | $-\frac{d[A]}{dt} = k[A]$   | $\ln[A] = \ln[A]_0 - kt$                    | $\frac{\ln 2}{k}$       |
| <b>2</b>       | $-\frac{d[A]}{dt} = k[A]^2$ | $\frac{1}{[A]} = \frac{1}{[A]_0} + kt$      | $\frac{1}{k[A]_0}$      |
| <b>3</b>       | $-\frac{d[A]}{dt} = k[A]^3$ | $\frac{1}{[A]^2} = \frac{1}{[A]_0^2} + 2kt$ | $\frac{3}{2k[A]_0^2}$   |

- **Activation energy ( $E_a$ )**
  - 1) The minimum energy required to initiate a chemical reaction.
  - 2) The difference between the energy of the starting molecule and the highest energy along the reaction pathway.
  - 3) The rate depends on the magnitude of  $E_a$ , generally, the speed of reaction would increase with the decrease of the value of  $E_a$ .
- **Activated complex (transition state):** the molecule having the arrangement of atoms shown at the top of the barrier.



- **The Arrhenius equation:**

$$k = Ae^{-\frac{E_a}{RT}}$$

$$\ln k = -\frac{E_a}{RT} + \ln A \quad \ln k = \left(-\frac{E_a}{R}\right) \frac{1}{T} + \ln A$$

$$\ln k_2 - \ln k_1 = \left(-\frac{E_a}{RT_2} + \ln A\right) - \left(-\frac{E_a}{RT_1} + \ln A\right)$$

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right) = \frac{E_a}{R} \left(\frac{T_2 - T_1}{T_1 \cdot T_2}\right)$$

k: rate constant;

$E_a$ : activation energy;

R: gas constant;

T: absolute temperature;

A: frequency factor, related to the frequency of effective collision;

- **Reaction mechanism:** the steps by which a reaction occurs.
  - 1) **Elementary reaction:** a chemical reaction in which one or more chemical species react directly to form products in a single reaction step and with a single transition state.  
 $\text{CO} + \text{NO}_2 \rightarrow \text{CO}_2 + \text{NO}$   
 $\text{rate} = k[\text{CO}][\text{NO}_2]$
  - 2) **Multistep mechanism:** the reaction appears to proceed in a sequence of elementary reactions.  
 $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$   
 $\text{rate} = k[\text{H}_2][\text{Br}_2]^{1/2}$ 
    - ①  $\text{Br}_2 \rightleftharpoons 2\text{Br}$  (fast)
    - ②  $\text{Br} + \text{H}_2 \rightarrow \text{HBr} + \text{H}$  (slow) → Rate determining step
    - ③  $\text{H} + \text{Br}_2 \rightarrow \text{HBr} + \text{Br}$  (fast)
- **Molecularity:** the number of molecules that participate as reactants in an elementary reaction.

- **Intermediate:** the substance that formed in one elementary reaction and consumed in the next.

- **Rate Laws:**

- 1) The rate law for *elementary reaction* is based directly on its molecularity.

| Molecularity | Elementary Reaction                         | Rate Law                   |
|--------------|---------------------------------------------|----------------------------|
| Unimolecular | $A \longrightarrow \text{products}$         | $\text{Rate} = k[A]$       |
| Bimolecular  | $A + A \longrightarrow \text{products}$     | $\text{Rate} = k[A]^2$     |
| Bimolecular  | $A + B \longrightarrow \text{products}$     | $\text{Rate} = k[A][B]$    |
| Termolecular | $A + A + A \longrightarrow \text{products}$ | $\text{Rate} = k[A]^3$     |
| Termolecular | $A + A + B \longrightarrow \text{products}$ | $\text{Rate} = k[A]^2[B]$  |
| Termolecular | $A + B + C \longrightarrow \text{products}$ | $\text{Rate} = k[A][B][C]$ |

- 2) The *slowest step (rate-determining step/rate-limiting step)* in a *multistep reaction* determines the overall rate.

- **Catalyst:** <sup>alter A or E<sub>s</sub></sup> a substance that changes the speed of a chemical reaction without undergoing a permanent chemical change itself, it lowers the overall activation energy for a chemical reaction.
  - 1) **Homogeneous catalyst:** a catalyst that is present in the same phase as the reactants in a reaction mixture;
  - 2) **Heterogeneous catalyst:** a catalyst that exists in a phase different from the phase of the reactant molecules, usually as a solid in contact with either gaseous reactants or reactants in a liquid solution.
  - 3) **Enzyme:** a large number of marvelously efficient biological catalysts.
- **Active site:** the reaction any given enzyme catalyzes takes place at a specific location in the enzyme.
- **Substrate:** the substances that react at this site.
- **Lock-and-key model:** the substrate is pictured as fitting neatly into the active site, much like a key fits into a lock.
- **Enzyme-substrate complex:** the combination of enzyme and substrate.
- **Enzyme inhibitor:** the activity of an enzyme is destroyed if some molecule other than the substrate specific to that enzyme binds to the active site and blocks entry of the substrate.